React Native Assessment

Stack: React Native + Expo (Expo Router, TypeScript)

Task: Build "ShopLite", a product browsing app

Build a mobile app using the public API **https://dummyjson.com** (`/products`, `/products/search?q=`, `/products/{id}`).

Requirements

1. Navigation

- Bottom tabs (Home, Favorites) and a product details screen at `/product/[id]`
- The deep link `shoplite://product/5` must open that product directly, with a working back action

2. Product list (Home)

- Put all fetching, pagination, refresh and error logic in a reusable custom hook `useInfiniteProducts`
- Infinite scroll with `FlatList` (20 items per page) and pull-to-refresh. Prevent duplicate page requests when the end of the list is reached repeatedly, and never render duplicate items.
- Search with a 400 ms debounce using a custom `useDebounce` hook
- Sort by price or rating, ascending or descending (`sortBy` and `order` query params). Changing the search or sort resets the list and pagination.
- Cancel outdated requests with `AbortController` so an older API response never overwrites a newer one
- Skeleton loaders, plus empty, error (with Retry) and "loading more" states
- Memoized list items and stable callbacks so scrolling stays smooth with 100+ items

3. Product details

- Render instantly using the product data already loaded in the list, then refresh it from the API in the background
- Horizontal paging image carousel with page indicator dots
- Title, price, discounted price (calculated), rating, stock status and a favorite toggle in the header

4. Favorites

- Global state using Context + `useReducer` or Zustand
- Persisted with `AsyncStorage` and restored on launch without a flash of empty content
- Favorites screen uses a `SectionList` grouped by category
- Removing a favorite shows an "Undo" snackbar for 4 seconds that restores the item in its original position
- Favorite status stays in sync across Home, Details and Favorites screens

5. Offline support

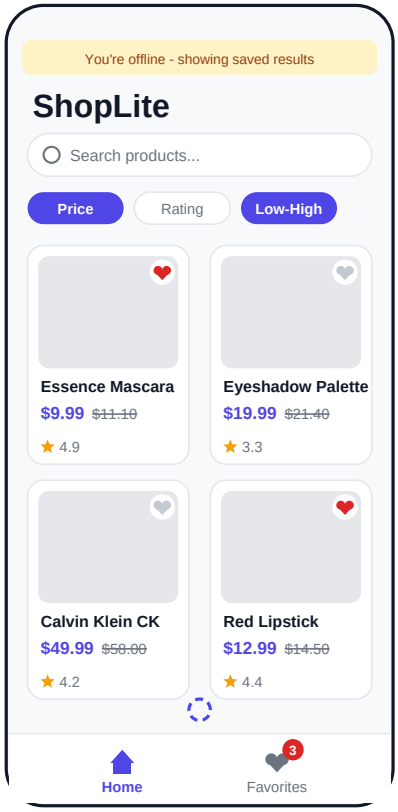
- Detect connectivity with `@react-native-community/netinfo` and show an offline banner
- Cache the first page of products, and show it when the app starts offline
- Retry automatically when the connection comes back

6. Testing

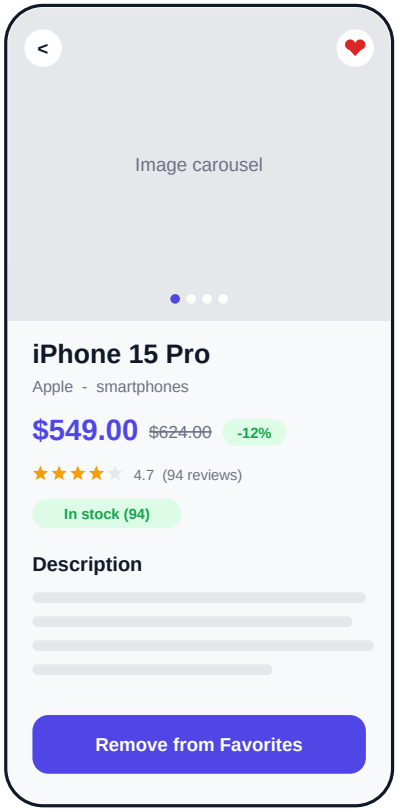
- Write Jest unit tests for the favorites reducer/store and the `useDebounce` hook

UI Design

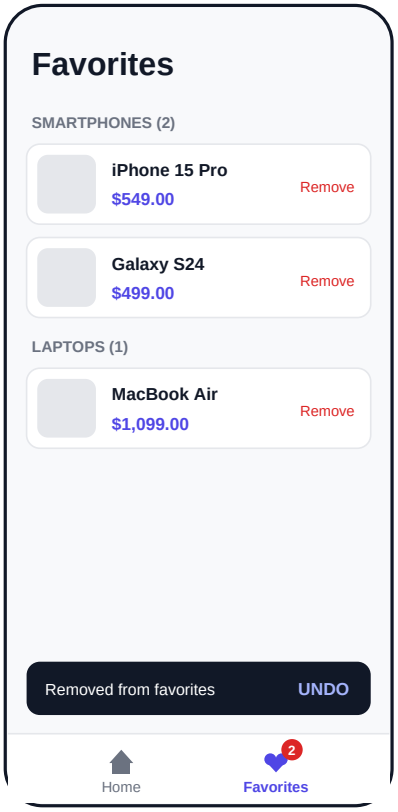
The app should look clean and modern, close to the wireframes below. Exact pixels are not required, but layout, hierarchy and the design tokens must be followed consistently across all screens.



Home



Product Details



Favorites

Design tokens

Token	Value	Token	Value
Primary	#4F46E5	Background	#F8F9FB
Text	#111827	Muted text	#6B7280
Success	#16A34A	Warning / rating	#F59E0B
Danger	#DC2626	Border / skeleton	#E5E7EB
Spacing scale	4, 8, 12, 16, 24	Radius	Cards 12, buttons 10, chips and pills full
Font sizes	Title 24, heading 18, body 14, caption 12	Touch targets	At least 44 x 44

Store the tokens in a single theme file and use them everywhere. No hard-coded colors inside components.

Home screen

- Large "ShopLite" title, then a rounded search bar with a search icon and a clear (x) button when text is entered
- A row of chips under the search bar for sort field (Price, Rating) and order (Low-High, High-Low). The active chip is filled with the primary color.
- Products in a 2-column grid of white cards: square image, heart icon overlaid at the top-right of the image, title (max 2 lines), discounted price in bold primary color next to the original price struck through, and a star with the rating
- Initial load shows 6 grey skeleton cards with the same shape as real cards. A small spinner appears at the bottom while the next page loads.
- Offline: an amber banner at the top saying the user is offline and seeing saved results

- Empty search: centered icon with "No products found for '<query>'". Error: message with a primary Retry button.

Product details screen

- Full-width image carousel covering about 40% of the screen, with page dots at the bottom. Floating round back button at top-left and heart button at top-right.
- Title (heading), brand and category in muted text
- Price row: discounted price large in primary color, original price struck through, and a green discount badge (for example -12%)
- Five-star rating row with the numeric value
- Stock pill: green "In stock (n)" when 10 or more, amber "Only n left" below 10, red "Out of stock" at 0
- Description section, and a full-width primary button at the bottom that reads "Add to Favorites" or "Remove from Favorites"

Favorites screen

- Category section headers in small uppercase muted text with the item count, for example SMARTPHONES (2)
- Each row: rounded thumbnail, title, price in primary color and a red Remove action on the right
- After removal, a dark snackbar at the bottom reads "Removed from favorites" with an UNDO action
- Empty state: centered heart icon, "No favorites yet" and a "Browse products" button that goes to Home

Tab bar

- Home and Favorites tabs with icons and labels. Active tab in primary color, inactive in muted grey.
- A red badge on the Favorites tab showing the number of saved items

Expected Output

The submitted app must demonstrate the following behaviour when run with `npx expo start` on Android or iOS:

Scenario	Expected result
App launch	Skeleton cards appear, then a 2-column grid of 20 products. Saved favorites are shown correctly with no flicker.
Scroll to the end	Next 20 products load with a bottom spinner. No duplicate items and no duplicate requests.
Pull down	List refreshes from page 1
Type "phone" quickly	Only one request is made after typing stops (400 ms). Results show only matching products. Clearing the search restores the full list.
Select Price, Low-High	List resets and shows products from cheapest to most expensive
Tap a product	Details screen opens instantly with title, price and image, then updates with full data
Tap the heart	Heart state and the tab badge update on every screen immediately
Close and reopen the app	Favorites are still there
Remove a favorite, tap UNDO	Item returns to the same position in the same section
Turn on airplane mode, restart	Offline banner appears and the cached first page is shown. Turning the network back on hides the banner and reloads data automatically.
Open shoplite://product/5	App opens directly on product 5 and the back button returns to Home
API failure	Error message with Retry. Retry recovers without restarting the app.
Run npm test	All unit tests pass

Deliverables

- GitHub repository link
- README with setup steps, folder structure, state management choice and any known limitations
- A 2 to 3 minute screen recording showing every scenario in the Expected Output table
- The app must run without crashes, red screens or TypeScript errors